

WASP-39b

R-1713

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-39_140513 · created 2026-08-02T15:45:28+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456790.679 +/- 0.025 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.191 +/- 0.045
Transit depth (Rp/Rs)^2	3.65 +/- 1.74 [%]
Orbital Inclination (inc)	89.95 +/- 1.74
Airmass coefficient 1 (a1)	1.202 +/- 0.035
Airmass coefficient 2 (a2)	-0.147 +/- 0.056
Scatter in the residuals of the lightcurve fit is	2.9 %
Best Comparison Star	None
Optimal Aperture	3.2
Optimal Annulus	8.69
Transit Duration (day)	0.082 +/- 0.024

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.191 ± 0.045 vs 0.1457 (deviation 1.01σ)
Duration fit vs published	0.082 d vs 0.12 d (deviation 1.58σ)
Mid-time O–C	-36.9 min
Depth SNR	4.2
Residual scatter	2.9 %

Light curve

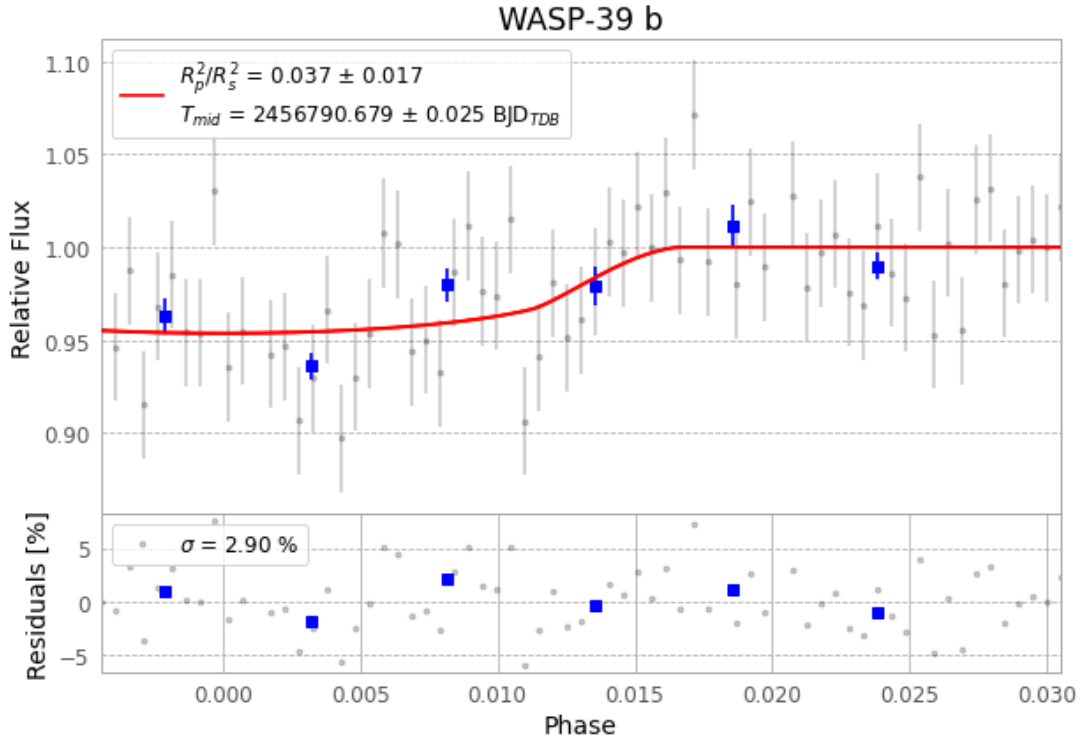


Figure 1 — FinalLightCurve_WASP-39 b_2014-05-12.png (SHA-256 4689ae2276d27b4c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [209, 203], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 f6322775dba7a92b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

69 FITS frames (45 MB), WASP-39140513034212.FITS → WASP-39140513070613.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-39-140513.log (6ae7e5eef4c2...), FinalLightCurve_WASP-39 b_2014-05-12.png (4689ae2276d2...), FinalLightCurve_WASP-39 b_2014-05-12.csv (7dfab11b4b68...)

Compute resources

Wall time 130.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 15:45Z.